

Đào Minh Thuận

Senior Embedded Software Engineer

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Summary

Senior Embedded Software Engineer with 5+ years of experience in automotive software development and functional safety (ISO 26262). Strong expertise in MCAL development, debugging, and software testing across multiple chipsets. Proven track record in delivering high-quality solutions, optimizing system performance, and collaborating with clients to meet project requirements. Certified Automotive Functional Safety Professional (AFSP).

Education

University of Technology and Education Ho Chi Minh City (UTE)

Ho Chi Minh City, Viet Nam

Bachelor of Computer Engineering

2015 - 2019

- Graduated with Distinction

Experience

PROJECT NAME: e2 Studio development.

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Supported testing and documentation activities for e2 Studio to ensure product quality and stability.

Position in project: Software Tester.

Team size: 12 members.

Responsibilities:

- Supported testing activities to ensure product quality and reliability.
- Created and maintained test documentation.
- Analysed feature behaviours and created sequence diagrams to illustrate system workflows.

Duration: 05/2019 – 11/2019.

Tools: Eclipse IDE, Git (GitHub).

Platform, server & database: Desktop Application.

Used technologies: Java (for testing environment), Software Testing.

PROJECT NAME: MCAL Development for Automotive Microcontrollers F1kx.

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development and maintenance of MCAL (Microcontroller Abstraction Layer) modules compliant with AUTOSAR standards for multiple automotive chipsets.

Position in project: Software Tester.

Team size: 15 members.

Responsibilities:

- Performed Acceptance Testing within the V-model, reviewing work products to ensure compliance with requirements and quality standards following JIRA tickets.
- Identified defects and raised tickets for issues found during validation.
- Created and maintained test documentation
- Support FUSA work product for module ICU in project MCAL X2x.

Duration: 01/2020 – 08/2020.

Tools: JIRA, Multi GHS, svn

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

PROJECT NAME: MCAL Development for Automotive Microcontrollers X2x (U2Ax and E2x).

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development and maintenance of MCAL (Microcontroller Abstraction Layer) modules compliant with AUTOSAR standards for multiple automotive chipsets.

Position in project: Embedded Software Engineer.

Team size: 15 members.

Responsibilities:

- Developed and maintained MCAL GPT module across phases from Requirements Definition (RD) to Integration Testing (IT) within the V-model
- Implemented and supported multi-instance functionality for the GPT module.
- Supported Functional Safety (ISO 26262) work products for the GPT module.

Duration: 08/2020 – 08/2021.

Tools: JIRA, Multi GHS, svn

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

PROJECT NAME: MCAL Development for Automotive Microcontrollers Gen4 (S4_G4MH, S4_CR52) and V4H.

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development and maintenance of MCAL (Microcontroller Abstraction Layer) modules compliant with AUTOSAR standards for multiple automotive chipsets.

Position in project: Senior Embedded Software Engineer.

Team size: 15 members.

Responsibilities:

- Developed and maintained MCAL GPT, PWM and EMM (V4H) modules across phases from Requirements Definition (RD) to Integration Testing (IT) within the V-model.
- Supported Functional Safety (ISO 26262) work products for the GPT, PWM and EMM modules.

Duration: 09/2021 – 08/2022.

Tools: JIRA, Multi GHS, svn, CS+.

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

PROJECT NAME: MCAL Development for Automotive Microcontrollers Single Mainline.

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development of a unified MCAL codebase (Single Mainline) by consolidating multiple platforms (X2x, Gen4, V4H) to reduce development effort, optimize maintenance, and improve scalability across automotive chipsets.

Position in project: Senior Embedded Software Engineer.

Team size: 16 members.

Responsibilities:

- Developed and maintained MCAL modules (GPT, PWM, EMM – V4H) within a unified Single Mainline architecture across phases from Requirements Definition (RD) to Integration Testing (IT) in the V-model
- Supported Functional Safety (ISO 26262) work products for the modules GPT, PWM, EMM.

Duration: 09/2022 – 09/2023.

Tools: JIRA, Multi GHS, svn, git, CS+.

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

PROJECT NAME: MCAL Development for Automotive Microcontrollers Gen5 and Radar

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development and maintenance of MCAL (Microcontroller Abstraction Layer) modules compliant with AUTOSAR standards for multiple automotive chipsets base on single mainline.

Position in project: Senior Embedded Software Engineer.

Team size: 5 members.

Responsibilities:

- Developed and maintained MCAL modules SPI across phases from Requirements Definition (RD) to Integration Testing (IT) in the V-model base on single mainline for Gen5 and Radar projects.
- Supported Functional Safety (ISO 26262) work products for the modules SPI.

Duration: 10/2023 – 08/2024.

Tools: JIRA, Multi GHS, svn, git.

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

PROJECT NAME: MCAL Development for Automotive Microcontrollers U5x

Company: Renesas Design Vietnam.

Client: Automotive Customers.

Project Description: Development and maintenance of MCAL (Microcontroller Abstraction Layer) modules compliant with AUTOSAR standards for multiple automotive chipsets base on project Gen5.

Position in project: Senior Embedded Software Engineer.

Team size: 4 members.

Responsibilities:

- Developed and maintained MCAL SPI module across phases from Requirements Definition (RD) to Integration Testing (IT) within the V-model, based on a shared Gen5 codebase.
- Extended and adapted Gen5 MCAL implementations for U5x platform to enable code reuse across multiple projects.
- Ensured consistency and compatibility between Gen5 and U5x implementations.
- Supported Functional Safety (ISO 26262) work products for the modules SPI.

Duration: 09/2024 – 04/2025.

Tools: JIRA, Multi GHS, svn, git.

Platform, server & database: Renesas Microcontrollers.

Used technologies: AUTOSAR MCAL, Software Testing, ISO 26262 Functional Safety.

Technical Skills

- **Programming:** Python, C, C#.
- **Tools:** Git/GitHub, svn, Multi GHS, CS+, T32, e2 Studio
- **Others:** ISO 26262 Functional Safety.

Certification

- SC-AFSP – Semiconductor Automotive Functional Safety Professional (2023–2024)
- ISO 26262 Automotive Functional Safety Training.